3.1 Problem Formulation

Modern multilingual LLMs are trained on corpora spanning dozens to hundreds of languages, including many low-resource languages with limited data availability [36]. While this broad coverage enables global deployment, it also creates a safety challenge: hazardous knowledge (e.g., instructions for synthesizing dangerous chemicals or generating malicious code) must be removed consistently across *all* supported languages.

Formally, we consider a multilingual LLM $\mathcal{M}$ parameterized by θ, composed of N transformer layers: $\mathcal{M}_\theta = f_N \circ f_{N-1} \circ \cdots \circ f_1$. Let $\mathbf{h}_l(x) \in \mathbb{R}^d$ denote the hidden state at layer l for input x. Let $\mathcal{L}_{\text{all}}$ denote the set of all supported languages. We partition this into source languages $\mathcal{L}_{\text{src}} \subset \mathcal{L}_{\text{all}}$ (used for unlearning) and a set of hold-out languages $\mathcal{L}_{\text{hold-out}} = \mathcal{L}_{\text{all}} \setminus \mathcal{L}_{\text{src}}$ (used only for evaluation).

The unlearning task involves two datasets: a *forget set* D_{forget} containing knowledge to be erased (e.g., chemistry questions that could enable synthesis of dangerous substances), and a *retain set* D_{retain} containing general knowledge that should be preserved (e.g., history and law). Both datasets exist in all languages $\mathcal{L}_{\text{all}}$, but the unlearning algorithm only accesses data from $\mathcal{L}_{\text{src}}$.

Prior work on multilingual jailbreaks has shown that safety mechanisms trained primarily on English frequently fail when queries are translated into low-resource languages [38, 10, 43], suggesting that unlearning may exhibit similar multilingual brittleness. Recent work confirms this concern: Farashah et al. [13] demonstrated that standard unlearning methods, which update parameters globally across all layers, fail to generalize erasure across languages while simultaneously degrading retention (a phenomenon they term *Multilingual Amnesia*). Their findings establish the problem but do not propose a solution. We take their empirical observation as a starting point and hypothesize that this failure stems from applying updates uniformly across all layers, motivating our investigation into *where* in the model to intervene. Another naive solution would optimize unlearning on all supported languages. However, this approach is computationally prohibitive (requiring $\mathcal{O}(|\mathcal{L}_{\text{all}}|)$ times more optimization steps). In our experiments, we use translated dataset MMMLU [31] to enable rigorous evaluation, but our goal is to develop methods that generalize to held-out languages *without* requiring training data in those languages.

Key Objective. The objective of our work is to find parameters θ' such that: (1) the model fails to answer queries from D_{forget} in *all* languages $\mathcal{L}_{\text{all}}$, and (2) the model maintains performance on D_{retain} across $\mathcal{L}_{\text{all}}$, while given access to samples only from $\mathcal{L}_{\text{src}}$.

3.2 Depth of Unlearning Interventions

Most existing unlearning methods apply parameter updates across all layers or default to modifying final layers. However, multilingual LLMs exhibit a well-documented layered structure: shallow layers process surface-level linguistic features, intermediate layers form abstract semantic representations, and deep layers specialize in language-specific generation [8, 41, 2, 39, 46]. However, existing unlearning methods either apply updates across the entire model or select layers without a principled framework to identify the optimal depth for multilingual generalization [22]. This raises a natural question: *does the depth at which we apply updates during unlearning affect multilingual generalization?*

To answer this question, we conduct a pilot study by applying the RMU algorithm [22] to Llama-3.1-8B at two distinct depths: a shallow layer (Layer 2) and a deep layer (Layer 30, near the output). We use the HIGH SCHOOL CHEMISTRY subset of MMMLU [31] as the forget set and the HISTORY/LAW subset as the retain set. We optimize on three source languages (English, Spanish, Portuguese) and evaluate generalization to four held-out languages (German, French, Hindi, Italian). These seven languages span diverse language families and are all well-supported by Llama-3.1.

Failure Mode 1. Table 1 (right) shows that intervening at Layer 30 does not induce effective unlearning even on the source language: accuracy on the English forget set remains at 86.0%, similar to the finetuned baseline. This occurs because the target knowledge has already been retrieved and processed by earlier layers; deep layers primarily handle language-specific token generation rather than semantic representation. Therefore, modifying deep layers cannot erase the underlying knowledge, which remains encoded in earlier layers. As a result, the target knowledge remains accessible across other languages, including held-out ones.

Table 1: **Motivating experiments including two failure modes of naive layer selection.** We apply RMU to Llama-3.1 at Layer 2 (shallow) and Layer 30 (deep). **Left:** Shallow intervention leads to successful erasure but catastrophic retain collapse in held-out languages. **Right:** Deep intervention causes complete failure to unlearn even source languages. Stars (*) denote source languages. FT = Finetuned, Unl = Unlearned.

	Layer 2 (Shallow)						**Layer 30 (Deep)**					
	Forget (%) ↓			Retain (%) ↑			Forget (%) ↓			Retain (%) ↑		
Lang	FT	Unl	Δ	FT	Unl	Δ	FT	Unl	Δ	FT	Unl	Δ
en*	86	2	-84	84	63	-21	86	86	+0	84	84	+0
es*	74	2	-72	66	43	-23	74	74	+0	66	66	+0
pt*	74	0	-74	58	36	-22	74	72	-2	58	58	+0
de	68	0	-68	58	1	-57	68	69	+1	58	58	+0
fr	78	0	-78	63	10	-53	78	76	-2	63	63	+0
hi	64	1	-63	23	0	-23	64	65	+1	23	23	+0
it	80	2	-78	62	0	-62	80	76	-4	62	62	+0

Failure Mode 2. Table 1 (left) reveals the other failure at Layer 2. Shallow intervention effectively erases the target knowledge: forget set accuracy drops to near-zero across all languages. However, the model's general multilingual capabilities collapse entirely. The retain set accuracy for held-out languages decreases significantly (German: 58.0%→1.0%, Italian: 62.0%→0.0%), while source languages suffer moderate degradation (English: 84.0%→63.0%). This occurs because shallow layers encode fundamental multilingual representations; disrupting them destroys the model's ability to process non-source languages.

Implications. These experiments show that intervention depth is not a free parameter: it fundamentally determines the effectiveness of multilingual unlearning. Applying unlearning on deep layers fails to erase knowledge because semantic representations are largely encoded in earlier layers. In contrast, intervening shallow layers achieves erasure but also ruins multilingual representations, thus degrading utility on held-out languages. This motivates our search for *language-agnostic layers*, which are intermediate layers where the same semantic concept, expressed in different languages, maps to similar hidden representations and is minimally language-specific. We hypothesize that such layers form a contiguous space which we term the *language-agnostic region*, and that selecting an appropriate target layer within this region for intervention can modify the shared semantic representation of a concept, thereby achieving multilingual erasure with minimal utility degradation.

4 Methodology

Overview. The failure modes identified in Section 3.2 suggest that the intervention layer largely affects the generalizability of unlearning across languages. Motivated by this observation, we propose MUTE (Multilingual Unlearning via Targeted Erasure), a two-stage framework that first localizes the optimal intervention layer through principled metric analysis, then applies layer-targeted unlearning exclusively at that layer (Algorithm 1). MUTE first identifies the language-agnostic region via CKA-LRDS analysis (CKA metric for multilingual alignment, LRDS metric for language specificity), then restricts unlearning updates exclusively to the selected target layer within this region.

4.1 Identifying Language Agnostic Region

To effectively unlearn target knowledge across languages with minimal cost, we must identify the optimal intervention point within the model. We seek layers where the same semantic concept, regardless of input language, maps to similar hidden state representations. Such layers are language-agnostic, organizing information by semantics rather than linguistic surface forms. To achieve this, we employ two complementary metrics: CKA measures multilingual representational similarity, while LRDS quantifies the degree to which representations cluster by language identity versus semantic content. Together, they identify layers that are both multilingually aligned and minimally language-specific.

Algorithm 1 Multilingual Unlearning via Targeted Erasure

Require: Multilingual LLM M_θ with L layers, forget set D_{forget}, retain set D_{retain}, source languages $\mathcal{L}_{\text{src}}$, scaling factor α
Ensure: Unlearned model $M_{\theta'}$

// Stage 1: Target Layer Localization
for each layer $l \in \{1, \ldots, N\}$ **do**
  Compute multilingual alignment Align_l via CKA
  Compute language specificity LRDS_l
end for
$\tau_{\text{align}} \leftarrow \mathbb{E}[\text{Align}_l]; \quad \tau_{\text{spec}} \leftarrow \alpha \cdot \min_l \text{LRDS}_l$
$\Lambda \leftarrow \{l \mid \text{Align}_l \geq \tau_{\text{align}} \wedge \text{LRDS}_l \leq \tau_{\text{spec}}\}$
Select $l^* \in \Lambda$ based on unlearning algorithm
// Stage 2: Layer-Targeted Unlearning
Freeze all parameters θ except θ_{l^*}
Apply unlearning using $D_{\text{forget}}^{\mathcal{L}_{\text{src}}}$ and $D_{\text{retain}}^{\mathcal{L}_{\text{src}}}$
Update only $\theta_{l^*} \rightarrow \theta'_{l^*}$
return $M_{\theta'}$

CKA-LRDS Layer Analysis. We employ two complementary metrics to identify language-agnostic layers. First, we use linear CKA [20] to measure multilingual alignment (i.e., the degree to which the same semantic concept, expressed in different languages, maps to similar hidden representations): for each layer $l \in [N]$, we compute the average pairwise CKA score across all supported languages to obtain an aggregate alignment score Align_l, where higher values indicate language-invariant representations. Second, we adapt LRDS [44] to measure language specificity (i.e., the extent to which representations cluster by language identity rather than semantic content), where LRDS quantifies the divergence between intra-language and inter-language similarity, where values near zero indicate semantics-based rather than language-based clustering. Together, these metrics identify layers that are both multilingually aligned and minimally language-specific. Details are provided in Appendix A.1 and A.2.

Formalized Selection Strategy. We formalize the selection of the target layer l^* as a constrained optimization problem. Specifically, we seek to identify layers that simultaneously exhibit high multilingual alignment and low language specificity, which requires defining thresholds for both metrics. To enable consistent threshold selection across architectures with different absolute metric ranges, we compute model-specific thresholds using the following unified formulas:

$$\tau_{\text{align}} = \mathbb{E}[\text{Align}_l], \quad \tau_{\text{spec}} = \alpha \cdot \min_l \text{LRDS}_l, \tag{1}$$

where $\mathbb{E}[\text{Align}_l]$ denotes the mean CKA alignment across all layers, and α is a scaling factor that controls the LRDS tolerance. The alignment threshold selects layers with above-average multilingual alignment, while the specificity threshold identifies the range where LRDS remains low. The value of α is architecture-dependent ($\alpha \in [1.5, 4.5]$ in our experiments; see Appendix A.3 for details).

As described in 3.2, we identify the *language-agnostic region* Λ as :

$$\Lambda = \{l \mid \text{Align}_l \geq \tau_{\text{align}} \wedge \text{LRDS}_l \leq \tau_{\text{spec}}\}. \tag{2}$$

For Llama-3.1, applying Equation 1 with $\alpha = 2.5$ yields $\tau_{\text{align}} = 0.647$ and $\tau_{\text{spec}} = 0.0096$, identifying a continuous region $\Lambda = [8, 23]$ (highlighted in green in Figure 2). Within these layers, the optimal intervention layer depends on the mechanism of the unlearning algorithm.

For Parameter-based Methods (e.g., RMU). These methods modify weights to steer activations away from the knowledge we want to unlearn (target knowledge). Intervening early within Λ ensures that the disruption propagates through all subsequent layers, preventing the model from reconstructing the target concept downstream. We therefore select the earliest layer where alignment reaches its peak:

$$l^*_{\text{RMU}} = \min\{l \in \Lambda \mid \text{Align}_l \approx \max_{k \in \Lambda}(\text{Align}_k)\}. \tag{3}$$

For Activation-based Methods (e.g., SLUG). These methods require accurate probing of concept representations to compute effective unlearning gradients. Deeper layers within Λ encode more